Jichen Chai

Atlanta, GA | 9403261863 | jichenchai@gmail.com | jchai49@gatech.edu |Permanent Resident

Education

Georgia Institute of Technology | Atlanta, GA *August 2025 – Present*
B.S in Electrical Engineering | Thread: Robotics & Autonomous Systems | GPA 3.70 Expected Graduation, May 2027

Collin College | Plano, TX *August 2023 – May 2025*
Computer Science | Transfer with 53 Credit Hours | GPA 4.00

Research Interns

Quadrupedal Navigation SLAM | Georgia Institute of Technology, LIDAR *January 2026 – Present*

- **Autonomous Navigation:** Engineered collision-free navigation architectures on Unitree Go 2 quadrupeds; Developed and debugged the perception, navigation, motion planning and control models in C++ and Python.
- **Simulation & Testing:** Leveraged ROS 2, Gazebo, MuJoCo, and RVIZ to architect high-fidelity simulation environments.
- **Sim-to-Real Deployment:** Facilitated high-fidelity sim-to-real deployment on quadrupeds by fine-tuning control parameters; Developed integration and test automation for LiDAR/radar/depth cameras navigation.

Optical System Control & Computer Vision | University of Texas at Dallas *September 2023 – May 2025*

- **Real-Time Feedback Control**: Developed a robust model to analyze system-level nonlinear autonomy behaviors; Designed a vision-based feedback optimization framework (PID controller); Achieved millisecond-level synchronization and adaptive stability;
- **Software-hardware Integration:** Engineered hardware-software integration over electrical and control systems using C/C++/Python; Synchronized the detection and illumination arm by controlling electrical components such as piezo and NI-DAQ.
- **Computer Vision:** Applied image registration, deep-learning segmentation, and cell tracking for data analysis of biological data.

Technical Projects

Unmanned Aerial Vehicle (UAV) Modeling | Georgia Institute of Technology

- Designed a blimp capable of obstacle avoidance control in unpredictable environments using ROS 2, Gazebo, and PX 4.
- Assisted with flight testing and data collection, helped integrate sensors and cameras.

Programming Hardware/Software Systems | Georgia Institute of Technology

- Engineered a high-performance pathfinding algorithms using RISC-V Assembly/ C. Optimized for the ESP32 microcontroller through manual memory mapping and register-level tuning to squeeze out every bit of efficiency.
- Developed a memory-constrained C-based hash table library designed for maximum performance. Ensured rock-solid data integrity and low-level stability via rigorous register-level debugging and hardware-in-the-loop (HIL) testing.

VR Development | University of Texas at Dallas

- Engineered a VR data visualization suite using Unity and C#, enabling 3D cardiac cell visualization.

Skills

Robotics Frameworks: ROS 2, Gazebo (Physics Modeling), RVIZ, Isaac Sim, Unity, Unitree Go 2, Unreal, MuJoCo
Programming: Python (Automation/API Development), C, C++, C#, MATLAB, Java, RISC-V (Low-level optimization)
Software & Integration: GitHub (Version Control/CI-CD workflow), Linux/Ubuntu, Docker, LabVIEW
Hardware Interface: Piezo Actuator, Galvo Mirror, CMOS Sensors, NI Data Acquisition, PLC, Arduino, Raspberry Pi
Language: English, Chinese, Japanese

Publications

- Zhang, X., Chai, J. (co-first author), et al. (2025). Instantaneous Volumetric Light-sheet Imaging of Beating Heart. *Optica Biophotonics Congress: Optics in the Life Sciences*.
- Zhang, X., Chai, J., et al. (2025). Compressed Sensing Light Sheet Microscopy System and Method. *US Patent* (Docket No. 25027-P1).
- Saberigarakani, …, Chai, J., et al. (2025). Volumetric Imaging and Computation to Explore Contractile Function in Zebrafish Hearts. *Cell Reports Methods*.